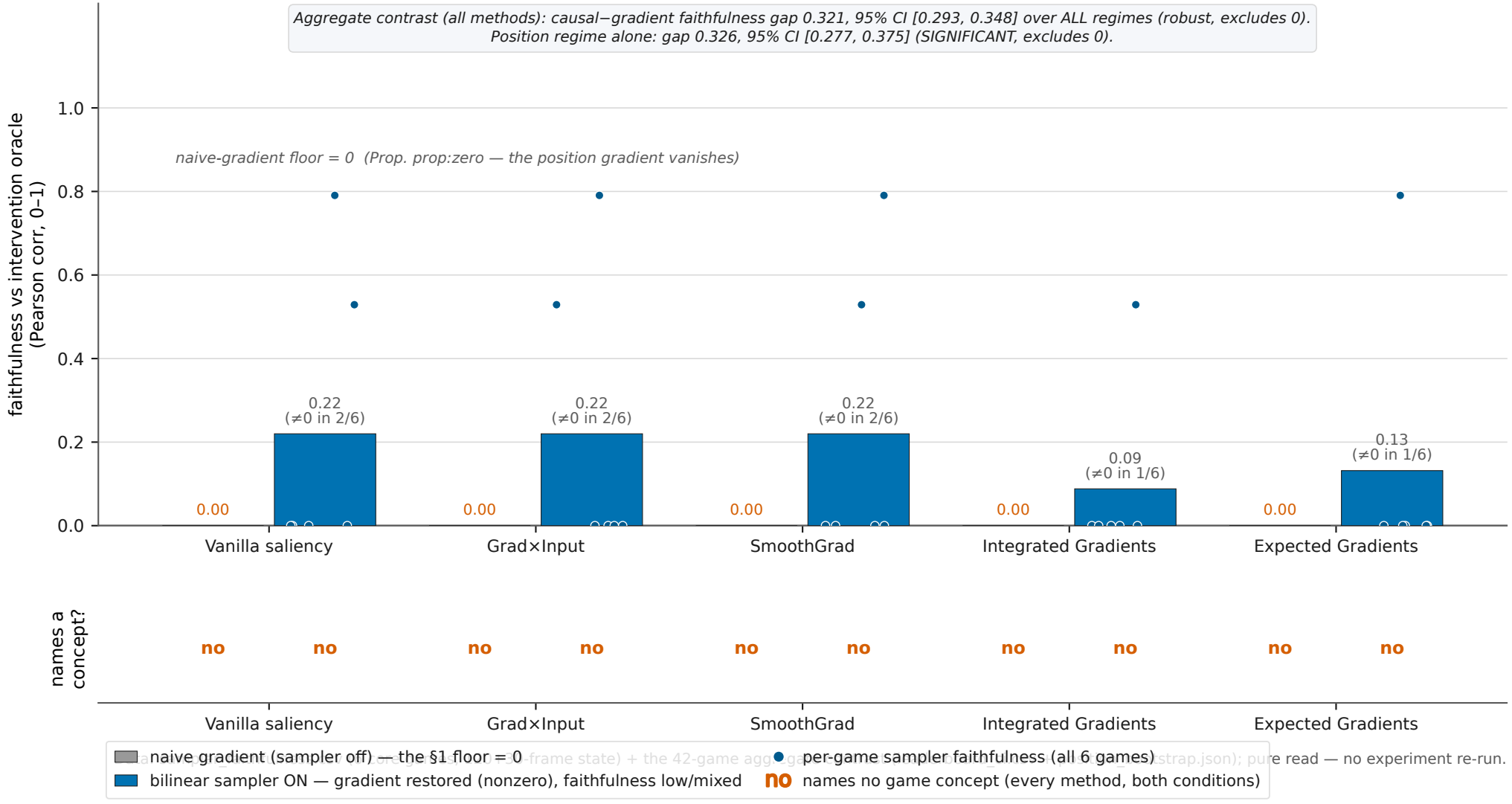
Aggregate contrast (all methods): causal−gradient faithfulness gap 0.321, 95% CI [0.293, 0.348] over ALL regimes (robust, excludes 0).
Position regime alone: gap 0.326, 95% CI [0.277, 0.375] (SIGNIFICANT, excludes 0).
faithfulness vs intervention oracle
(Pearson corr, 0–1)
1.0
0.8
0.6
0.4
0.2
0.0
naive-gradient floor = 0 (Prop. prop:zero — the position gradient vanishes)
0.00
0.22
(≠0 in 2/6)
0.00
0.22
(≠0 in 2/6)
0.00
0.22
(≠0 in 2/6)
0.00
0.09
(≠0 in 1/6)
0.00
0.13
(≠0 in 1/6)
Vanilla saliency
Grad×Input
SmoothGrad
Integrated Gradients
Expected Gradients
names a concept?
no
no
no
no
no
no
no
no
no
no
Vanilla saliency
Grad×Input
SmoothGrad
Integrated Gradients
Expected Gradients
naive gradient (sampler off) — the §1 floor = 0
bilinear sampler ON — gradient restored (nonzero), faithfulness low/mixed
per-game sampler faithfulness (all 6 games)
no
names no game concept (every method, both conditions)